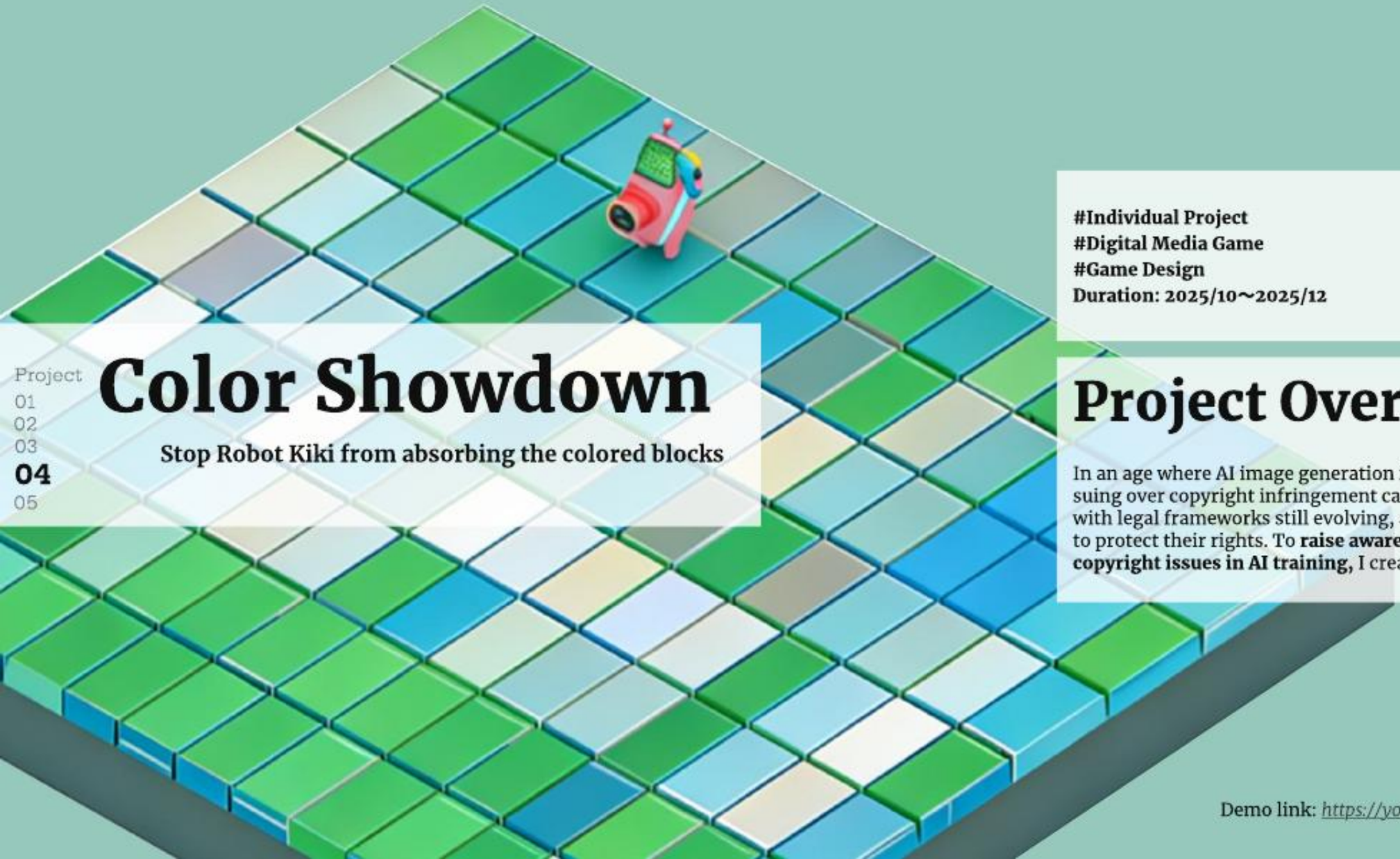


Project
01
02
03
04
05

Color Showdown

Stop Robot Kiki from absorbing the colored blocks

#Individual Project
#Digital Media Game
#Game Design
Duration: 2025/10～2025/12

Project Overview

In an age where AI image generation is widely accessible, more artists are suing over copyright infringement caused by AI-generated works. However, with legal frameworks still evolving, artists face a long and challenging fight to protect their rights. To **raise awareness about the potential for widespread copyright issues in AI training,** I created this game.

Demo link: https://youtu.be/yZNcYpi3fE4?si=G5boDpZ77h8zjFtt

Background



Hayao Miyazaki

AI-generated art is an insult to life itself.

Pete Docter

AI is the least impressive blah average of things.

Tony Bancroft

Please don't let AI steal my work.

Secondary Reseach

World's first AI image copyright infringement ruling



Works created through AI tools that reflect the author's creative expression are protected by copyright; If AI directly imitates others' works without original creative input, it constitutes infringement.



Listed among "Top 10 Events in China's Digital Economy and Legal Development (2024)."

History Research

Algorithmic Art

1960–70s



Generates geometric patterns with math and logic, controlling composition through algorithms
First fusion of art and programming, founding "computational creativity"

Procedural Generation

1990s



Uses texture algorithms and image stitching to build complex scenes in games and graphics
Generation evolves from artistic experiment to industrial application

Deep Learning & GAN

2010s



Trained on large datasets to generate photorealistic images and videos
Applications include deepfakes, image restoration, and style transfer

Text-to-Image Generation

2020s



Representative models: Stable Diffusion, DALL·E 2, Midjourney
Sparked copyright and ethical debates; significantly lowered the barrier to art creation

Multimodal Fusion

2024 - Present



Representative models: Sora (OpenAI), Runway Gen-3 Alpha
Text, image, video, and physics-consistent generation
AI generation trends toward real-time, immersive, and physically realistic outputs

Ideation

Game mechanic

Robot KIKI achieves victory by converting all non-target colors within the designated area to the target color. Players must select positions and strike sensors to turn the chosen area into a non-target color, thereby disrupting KIKI's operations. These gameplay mechanics mirror how AI image generation models, during training, cleanse and encode original artists' works without permission before learning from them. Humans, seeking to prevent this behavior, have developed interference tools designed to mislead the AI.

Art Style

Cartoon style



Game Target

"Stopping AI infringement is never easy."



Players must strategically choose positions
and hit sensors
repeatedly to block the robot's route.

Play Form

Single-player game



Game Progress

On the colored block board, KIKI attempts to convert all blocks to the target color. Players can use the mouse to select positions on the board and flip tiles. After flipping, a random contrasting color appears. Players must tap the sensor a specific number of times to generate the corresponding number of steps, locking the color in that square. This disrupts the robot's path planning, preventing it from converting the entire board to the target color.

Hand Interaction feature

The player taps the sensor downward



- 1 tap
- 2 taps
- 3 taps
- 4 taps
- 5 taps

Players can change the selected color block to the corresponding color **by tapping a different number of times.**

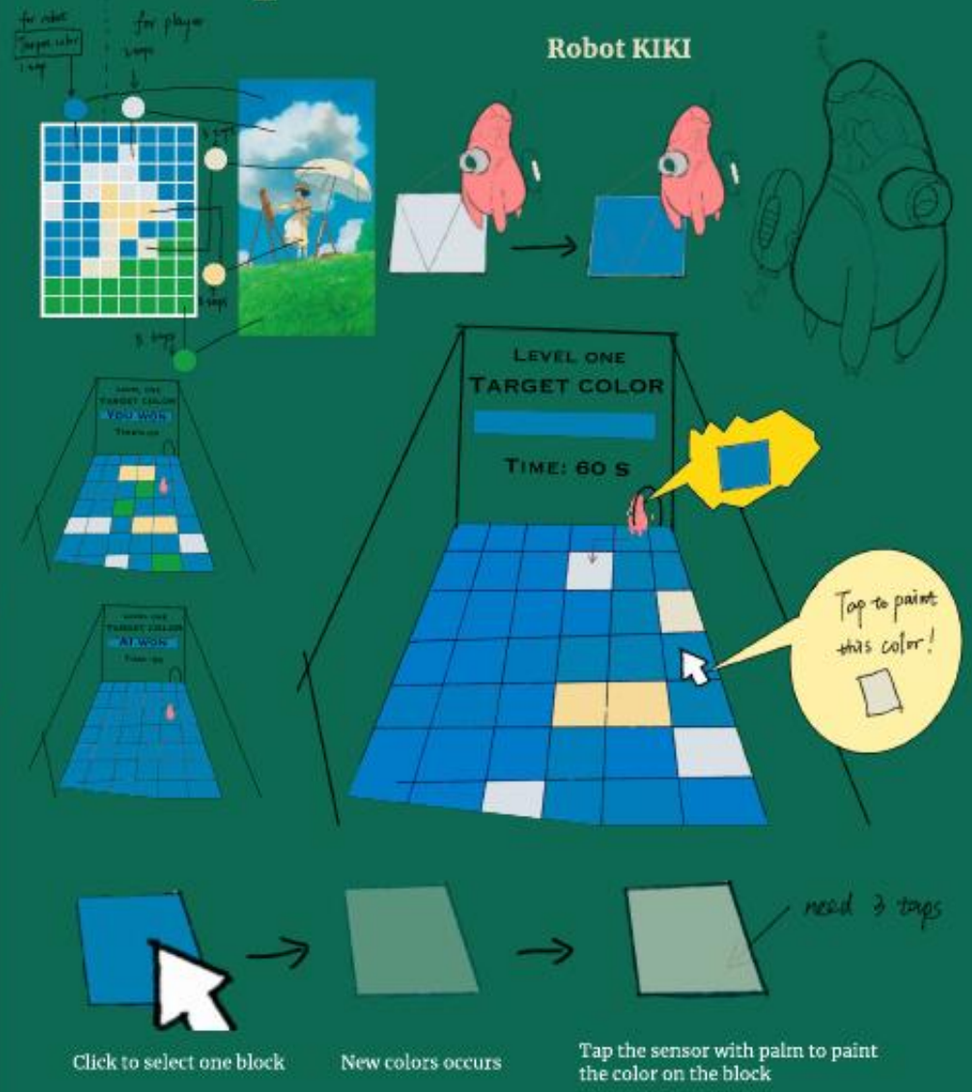
Some colors are so similar that they're hard to distinguish—a metaphor for how AI-generated content, after stealing artists' work, produces indistinguishable duplicates.

Game Flow

Turn-Based Gameplay

Level Selection	Game Setup	Robot Phase	Player Phase	Victory Check
Player chooses a level to begin the game	System generates a grid of colored tiles Displays robot opponent and target color Shows move/time limits	Robot Planning Robot autonomously plans its route Absorbs tiles and converts them to target color	Player Action Player flips tiles; colors change randomly Tap to lock tile color	Target reached → Player wins Time end → Robot wins
				

Concept



Development



TTP223 Touch Sensor Module

Detect external touch



Arduino Expansion Board

Connect modules more conveniently without breadboards



Arduino development board

Provide drivers

Coding

```
#define touchPin 2;

int currentState = 0;
int lastState = 0;

void setup() {
  Serial.begin(9600);

  pinMode(touchPin, INPUT);
}

void loop() {
  currentState = digitalRead(touchPin);

  if (currentState == HIGH && lastState == LOW) {

    Serial.println("tap");

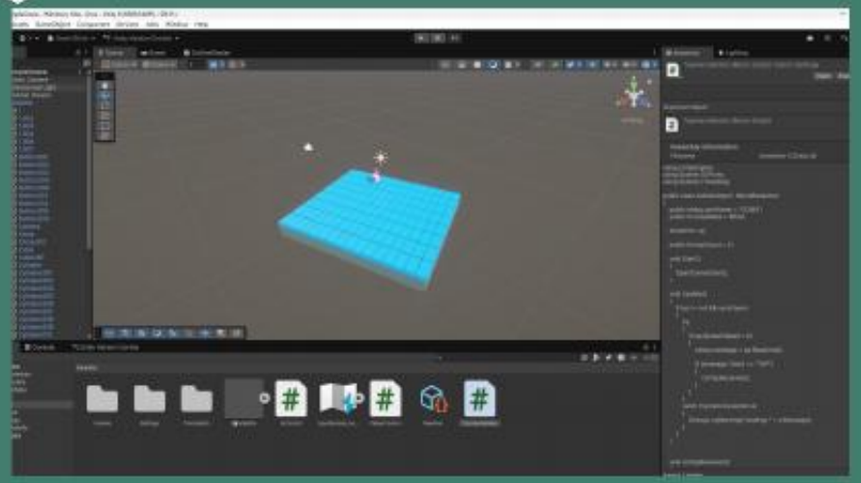
    delay(50);
  }

  lastState = currentState;
```



Computer Screen

Unity



Coding





Testing

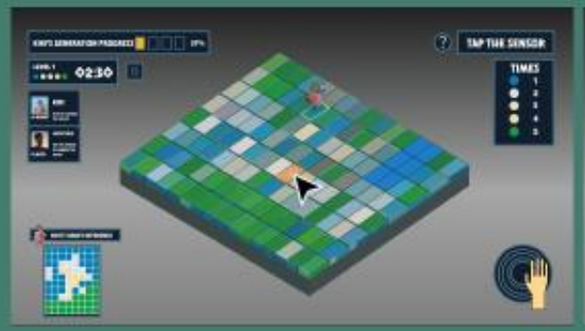


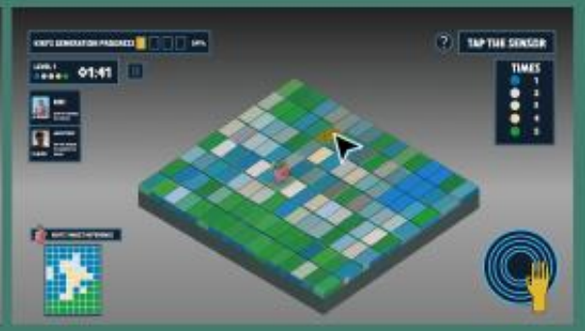
Character modeling process

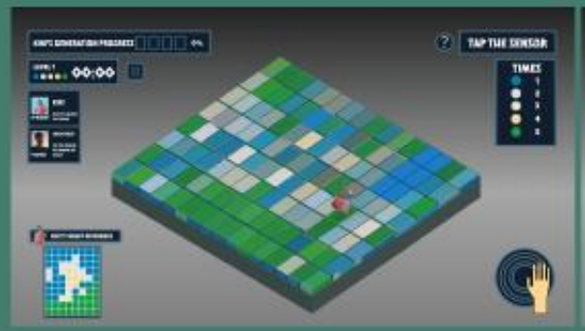


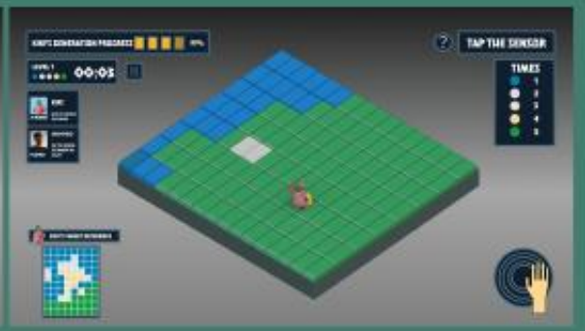


Game play







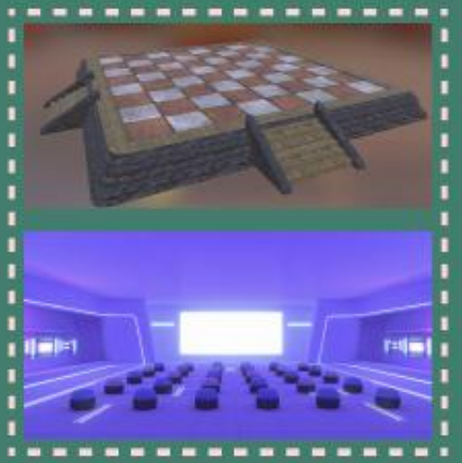






Game Assets





3D asset Used

Game UI







